

A Recursive Concentrator Structure with Applications to Self-Routing Switching Networks

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Abstract—A technique for constructing an N -input concentrator from two $(N/2)$ -input concentrators and a stage of 2×2 switching elements is described. Recursive decomposition of each of the $(N/2)$ -input concentrators yields a $\log_2 N$ -stage switching network similar in topology to the inverse omega network. The control algorithm for setting the switching elements is shown to be a simple logic function of the activity bits of the inputs. An iterative cell that combines the functions of the control logic and the 2×2 switching element can be used as a building block to realize the entire concentrator. Applications of the concentrator structure as an interconnection fabric in fast packet switching is explored. This leads to an efficient self-routing network, similar in topology to the transposed Batcher-banyan network.

I. INTRODUCTION

It is well known [1, 2] that the combination of a running adder and an inverse omega network (also known as the transposed omega network or the reverse banyan network) can perform the functions of a concentrator. In this approach, the running adder computes a set of new destination tags for the active inputs and the individual bits of the tags are used to control the setting of the 2×2 switching elements in the inverse omega network. It can be shown [3, 4] that the sequence of tags formed by the running adder defines a permutation that can be realized by the inverse omega network.¹

A sequence of concentrators can be employed to realize a completely specified permutation [5] as follows: First, inputs with most significant bit (MSB) equal to zero are treated as active and concentrated to the top half of the network while the remaining inputs (with MSB = 1) are automatically directed to the bottom half. This operation sorts the inputs into two groups based on the MSB of the destination tags. By continuing this binary radix sorting process, based on the other bits of the tags, one can complete the sorting operation which in turn realizes the required permutation.

In this paper we take a different approach to demonstrate the concentration property of the inverse omega network. We first show a technique that employs a stage of 2×2 switching elements and two $(N/2)$ -input concentrators to construct an N -input concentrator. This technique applied

recursively yields a $\log_2 N$ -stage network that is topologically similar² to the inverse omega network. It also leads to a simple iterative cell design that incorporates both the switching and the control functions. The entire concentration network can then be constructed using these iterative cells.

We next consider the application of concentrators in the design of fast packet switches. The binary radix sorting network mentioned above can be used only when all the packets at the switch inputs are active and when there are no duplicated destination requests. We show that the deployment of an additional front-end concentrator can alleviate the problem caused by inactive inputs. The combination of the front-end concentrator and the concentrator-based binary radix sorter yields a network similar in topology to the transposed Batcher-banyan network.

II. RECURSIVE CONSTRUCTION OF CONCENTRATORS

The traditional concentrator has M inputs and $N < M$ outputs, and interconnects a given set of N inputs to the specified N outputs. In this paper, however, we look at a concentrator slightly differently. As shown in Fig. 1, the concentrator is an $N \times N$ network. Its function is to route all the active inputs (whatever may be the definition of activity) to the top end of the network outputs.

An N -input concentrator can be constructed from two $(N/2)$ -input concentrators and a switching stage as shown in Fig. 2. In this method, the states of the front-end 2×2 switches are set so that one-half of the active inputs is routed to the upper concentrator. This action forces the other half to automatically trickle down to the lower concentrator. (If there are an odd number of active inputs, the upper concentrator gets one more active input than the other.) The outputs from the two $(N/2)$ -input concentrators are *interleaved* to obtain the final concentrated output. By decomposing each of the $(N/2)$ -input concentrators recursively, we obtain a network consisting of $\log_2 N$ stages of 2×2 switching elements.

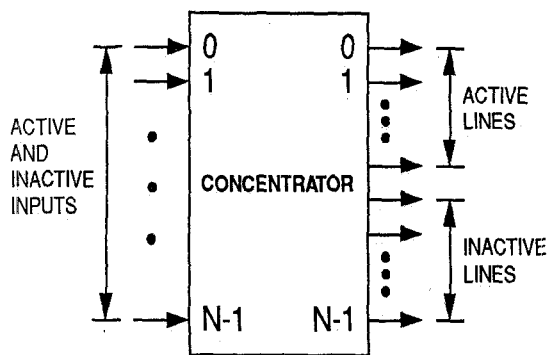
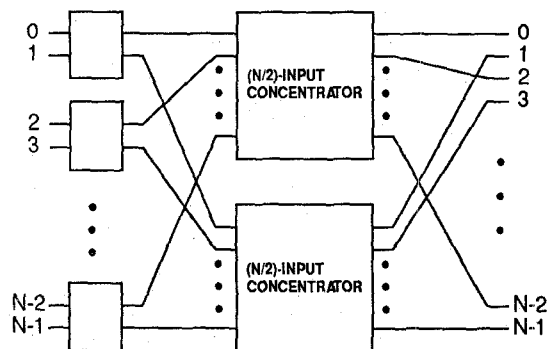
A simple technique that yields the required equal distribution is to set the 2×2 switches, starting from the top, so that the active inputs are directed alternately to the upper

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¹It is simpler to show that the inverse of this permutation represents a bitonic sequence and can therefore be realized by an omega (or, banyan) network [4, 7].

²We use the term "topologically similar" to emphasize the fact that the networks described in this paper are constructed out of iterative cells, each cell consisting of a 2×2 switching element with the control function built in. If we ignore the control function part of these cells in considering the topological equivalence, then the term "topologically identical" would be more appropriate.

Fig. 1. An $N \times N$ concentrator.Fig. 2. Recursive construction of an N -input concentrator.

and lower concentrators. The following logic accomplishes this task:

- (a) If both the inputs at a switching element are either active or inactive, then the switch can be set arbitrarily.
- (b) If one of the inputs is active, then it is routed to the lower concentrator if the previous active input was directed to the upper concentrator, and vice versa.

An iterative cell for performing this control function is shown in Fig. 3. In this diagram, a_i and b_i denote the activity bits of the two inputs, and c_i denotes the state of the previous active input. It is at logic zero if the previous active input was routed to the lower concentrator; otherwise it is at logic one. The cascade output to the next cell is denoted by c_{i+1} . The state of the 2×2 switching element is controlled by d_i . A logic zero on this control line sets the switch to the straight-through state while a logic one sets it to the crossed state.

The decomposed complete concentrator for eight inputs, based on the iterative cell of Fig. 3, is shown in Fig. 4. The sequence of output interleaving implied by the recursion corresponds to a bit-reversal permutation between the outputs of the final switching stage and the network outputs. In fact the topology of this network is similar to that of the "R-network" of Parker [6]. It is shown in [6] that the R-network is identical in topology to the inverse omega network. We can demonstrate this equivalence by swapping the positions of the switching elements 1 and 2 in the last stage. The resulting network structure is shown in Fig. 5. It has $\log_2 N$ stages, each stage consisting of an exchange permutation and an unshuffle permutation.

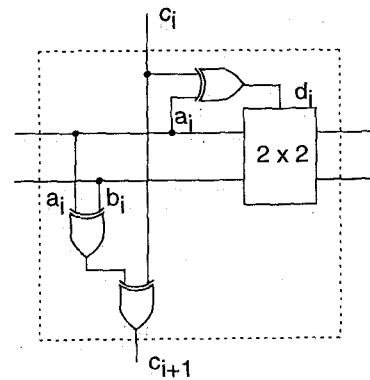


Fig. 3. An iterative cell for constructing the concentrator.

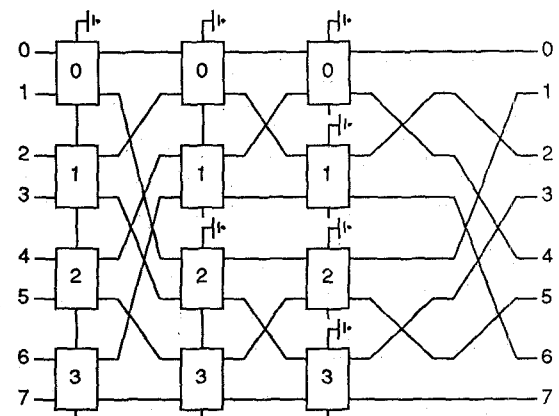


Fig. 4. An 8-input concentrator constructed from iterative cells.

As mentioned above, the inverse omega network is capable of performing the concentration function. We have given a simple explanation of this property through the recursive construction technique. The function of the running adder has been incorporated into the iterative cell. The structure of Fig. 4 may be viewed as a distributed implementation of this function.

III. APPLICATIONS TO SELF-ROUTING SWITCHING SWITCHES

A concentrator can serve as a building block in constructing a binary radix sorting network as follows. First, we distribute the inputs into two groups based on the MSB of the destination tags. Treating the inputs with MSB of the tags equal to zero as active and concentrating them accomplishes this task. Then, each of these two groups is distributed again based on the next bit of the tags. We continue this process till the complete list is sorted. The resulting network structure for the case of eight inputs is shown in Fig. 6. It may be recognized as topologically similar to the transposed Batcher sorting network.

A sorting network can perform the switching function only if the destination tag sequence represents a completely specified permutation. Additional steps would be necessary to realize an input-output connection pattern with inactive

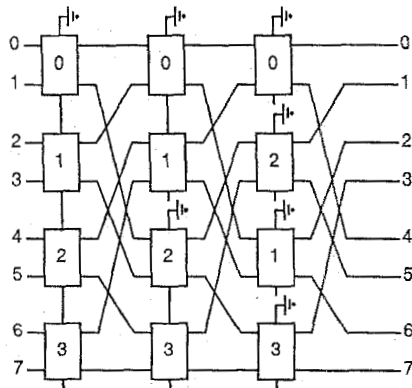


Fig. 5. The 8-input concentrator of Fig. 4 redrawn to demonstrate its topological similarity to the inverse omega network.

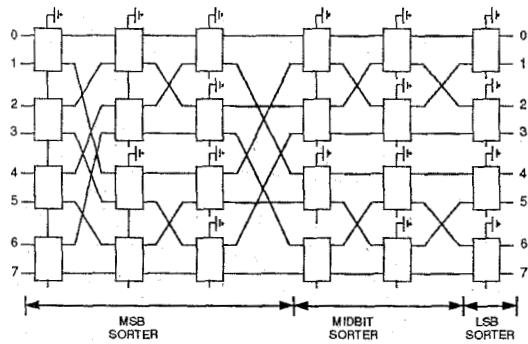


Fig. 6. An 8-input binary radix sorting network based on concentrators.

inputs. If sorting is performed by the Batcher network, which accomplishes the sorting of a long list by merging sorted sub-lists, a subsequent omega network can complete the required routing. This is the Batcher-banyan network [1,7].

The binary radix sorting system described above performs sorting by distribution. As illustrated in Fig. 7, a front-end concentration of the active inputs is necessary to route an incompletely specified permutation with this network. To see why the binary radix sorting network functions properly in this case, examine the distribution process (Fig. 2) again in the first stage of the MSB sorter (Fig. 6), for example. Except possibly for one, all the 2×2 switching elements in this stage have both their inputs either active or inactive. Starting from the top, inputs with MSB of the tags equal to zero are routed alternately to the upper and lower groups. This implies, starting from the top again, inputs with MSB of the tags equal to one are routed alternately to the lower and upper groups. The state of the one switching element with both active and inactive inputs is determined by the upper input, which is necessarily active, along with the routing status of the previous active input. Thus the iterative cell depicted in Fig. 3 functions properly even in the presence of inactive inputs. Switching elements with both the inputs inactive can obviously be set to an arbitrary state. It can be easily verified that this argument is also applicable to the distribution function performed by

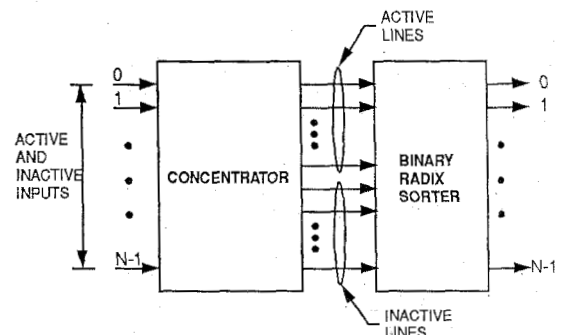


Fig. 7. Network structure for realizing incompletely specified permutations.

the other stages.

IV. CONCLUSION

We demonstrated the concentration capability of the inverse omega network starting from a recursive decomposition of the basic concentrating network. This technique allows us to define an iterative cell that can be used as a building block in constructing the concentrator. It also results in a distributed implementation of the running adder network. The traditional concentrator consisting of the running adder and the inverse omega networks requires the appending of additional routing tags to the original destination addresses whereas no such modifications are necessary with our approach.

Since the Batcher sorting network and the Batcher-banyan switching network can realize completely and incompletely specified permutations, respectively, it can be argued that the transposed versions of these networks should possess comparable capabilities, although the algorithms to set the individual switching elements in these networks are not obvious. Our derivation of these networks starting from the concentrator-based binary radix sorter clearly delineates these algorithms.

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